

AI Project Portfolio and Research Highlights

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Project Title

Judicial Bottleneck Diagnosis and Explainable Case Outcome Prediction in Tanzania Using SEMDR-Lite + ADAPT

Project Overview

This project applies Artificial Intelligence, Machine Learning, Natural Language Processing, and Explainable AI to support judicial analytics and evidence-based legal reform in Tanzania. The research addresses persistent challenges in judicial systems, including procedural delay, backlog accumulation, fragmented digital records, and limited analytical visibility into case-flow dynamics. The project develops an integrated framework combining bottleneck diagnosis, case outcome prediction, and governance-oriented interpretability.

Research Problem

Judicial institutions in many low- and middle-income countries face delays and limited data-driven decision-support capacity. In Tanzania, these challenges are intensified by uneven digitization, fragmented registry-level data, and bilingual judicial documentation in Swahili and English. This creates a need for intelligent, explainable, and governance-aware AI tools that can support public-sector modernization without replacing human judgment.

Objectives

- Diagnose judicial bottlenecks using filing-to-judgment duration and case-flow indicators.
- Develop AI-assisted models for explainable case outcome prediction.
- Construct and analyze bilingual judicial datasets from Tanzania and Kenya.
- Evaluate cross-jurisdictional model transferability under low-resource conditions.
- Support responsible AI adoption for judicial reform and institutional decision-making.

Methodology and AI Framework

The project proposes a hybrid **SEMDR-Lite + ADAPT** framework that integrates precedent retrieval, semantic representation, adaptive reasoning, and explainable AI. The framework was designed as a decision-support system rather than an automated adjudication tool.

- Data Engineering:** Judicial data collection, digitization, metadata reconstruction, outcome harmonization, and bilingual alignment.
- Bottleneck Diagnosis:** Stratified statistical analysis and regression-based delay diagnostics.
- Prediction Model:** SEMDR-Lite semantic representation combined with machine learning classifiers.
- Adaptive Reasoning:** ADAPT-inspired reasoning structure for legal outcome interpretation.
- Explainability:** LIME and SHAP used to interpret model behavior and identify legally meaningful signals.

Key Results

Dataset	Accuracy	Macro-F1	ROC-AUC
Kenya Test Set	0.83	0.81	–
Tanzania Pilot Set	0.79	0.77	0.84

The results show that the proposed hybrid model achieved stable performance across jurisdictions and demonstrated practical potential for low-resource judicial analytics. Interpretability analysis confirmed that the model relied on legally meaningful procedural and evidentiary cues such as affidavit, evidence, dismissed, and allowed.

Innovation and Contribution

- Developed a governance-aware AI framework linking bottleneck diagnosis, outcome prediction, and institutional feedback.
- Applied explainable AI to legal decision-support in a low-resource East African judicial context.

- Demonstrated cross-jurisdictional transfer from Kenyan judicial data to Tanzanian pilot data.
- Positioned AI as a human-in-the-loop decision-support tool for backlog management, resource allocation, and reform evaluation.

Digital Governance and SDG Relevance

This project contributes to digital governance and public-sector innovation by showing how AI can support transparent, accountable, and evidence-based institutional reform. It aligns strongly with **SDG 16: Peace, Justice and Strong Institutions** by promoting access to justice, institutional effectiveness, and responsible use of emerging technologies.

Technical Skills Demonstrated

Artificial Intelligence	Machine Learning
Natural Language Processing	Explainable AI
Python Programming	Data Analysis
Semantic Retrieval	Legal Analytics
LIME and SHAP	Digital Governance

Repository and Work Samples

GitHub Repository: <https://github.com/selemanimaregesi-blip/Judicial-AI-Tanzania>

Selected implementation notebooks, experimental outputs, figures, and reproducibility materials are available in the project repository. Sensitive or restricted judicial data are excluded in accordance with ethical and governance considerations.

Development Impact Statement

The project demonstrates how AI and data-driven methods can strengthen public institutions, improve service delivery, and support evidence-based policymaking. By combining technical modeling with governance safeguards, the work contributes to responsible digital transformation and practical innovation for justice sector modernization.